AthenaK output for post-processing analysis

This document collects which AthenaK output variables are needed to perform a certain analysis in post-processing, e.g. the visualization of the evolution of kinetic and magnetic energy in the case of GRMHD simulations.

Ejecta and Poynting flux

Variables

Metric: `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, `adm_gzz`
Primitives: `dens`, `s00`, `temperature`, `velx`, `vely`, `velz`, `bcc1`, `bcc2`, `bcc3`
Gauge: `z4c_alpha`, `z4c_betax`, `z4c_betay`, `z4c_betaz`
Total: $\boxed{N_{\text{vars}} = 23}$

Output type

```
<output1>
file_type = sph
variable  = ...   # see variable list
dt        = 20.0
radius    = 400.0 # radii = 300.0 400.0 ...
```

Setup

Output time: 10M or 20M (the latter might already be too large)
Resolution: $(\theta, \phi)_{\text{low}} = (128, 256)$ or $(\theta, \phi)_{\text{high}} = (256, 512)$
Radii: Between 300M and 500M

Disk usage

- Low resolution - $N_{\text{vars}} \times N_\theta \times N_\phi \times 8$ bytes $= 23 \times 128 \times 256 \times 8 \approx \boxed{6.03\text{MB}(r\Delta t)^{-1}}$
- High resolution - $N_{\text{vars}} \times N_\theta \times N_\phi \times 8$ bytes $= 23 \times 256 \times 512 \times 8 \approx \boxed{24.12\text{MB}(r\Delta t)^{-1}}$
- Total (100ms or 20000M):

	10M	**20M**
Low resolution	≈ 12.06 GBr^{-1}	≈ 6.03 GBr^{-1}
High resolution	≈ 48.24 GBr^{-1}	≈ 24.12 GBr^{-1}

Note To save disk-usage, this can be paired with the output from **Neutrino luminosity and average energy density** and also **GRMHD butterfly diagrams**.

Neutrino luminosity and average energy density

Variables

M1 variables: `rad_m1_F`, `rad_m1_E`, `rad_m1_N`
Metric: `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, `adm_gzz`
Gauge: `z4c_alpha`, `z4c_betax`, `z4c_betay`, `z4c_betaz`
Total: $\boxed{N_{\text{vars}}} = N_{\text{vars,E/N}} \times N_{\text{species}} + \underbrace{N_{\text{vars,F}}}_{\text{3 vector components}} \times N_{\text{species}} + N_{\text{vars,metric/gauge}} = 2 \times 4 + 3 \times 4 + 10 = \boxed{30}$

Output type

```
<output2>
file_type = sph
variable  = ...   # see variable list
dt        = 20.0
radius    = 400.0 # radii = 300.0 400.0 ...
```

Setup
Output time: 10M or 20M (the latter might already be too large)
Resolution: $(\theta, \phi)_{\text{low}} = (128, 256)$ or $(\theta, \phi)_{\text{high}} = (256, 512)$
Radii: Between 300M and 500M

Disk usage

- Low resolution - $N_{\text{vars}} \times N_\theta \times N_\phi \times 8$ bytes $= 30 \times 128 \times 256 \times 8 \approx \boxed{7.86\text{MB}(r\Delta t)^{-1}}$
- High resolution - $N_{\text{vars}} \times N_\theta \times N_\phi \times 8$ bytes $= 30 \times 256 \times 512 \times 8 \approx \boxed{31.46\text{MB}(r\Delta t)^{-1}}$
- Total (100ms or 20000M):

	10M	**20M**
Low resolution	≈ 15.72 GBr^{-1}	≈ 7.86 GBr^{-1}
High resolution	≈ 62.92 GBr^{-1}	≈ 31.46 GBr^{-1}

Note To save disk-usage, this can be paired with the output from **Ejecta** and also **GRMHD butterfly diagrams**.

Tracers

Variables
Derived variables: `u_t`
M1 variables (fluid frame): `rad_m1_absF`, `rad_m1_e`
Primitives: `win_Vi`, `dens`, `temperature`, `s_00`
Total: $\boxed{N_{\text{vars}}} = N_{\text{vars,absF/e}} \times N_{\text{species}} + +N_{\text{vars,der./primitives}} = 2 \times 4 + 7 = \boxed{15}$

Output type

```
<output3>
file_type = sph
variable  = ...    # see variable list
dt        = 20.0
r_min     = 150.0
r_max     = 1000.0
r_spacing = linear
nradii    = 128
ntheta    = 128
```

Setup Output time: 20M might be too expensive, but 50M might already be too coarse Resolution: $(r, \theta, \phi)_{\text{low}} = (128, 128, 256)$ or $(r, \theta, \phi)_{\text{high}} = (128, 256, 512)$
Radii: 128 radii between the outer part of the disk and the domain is a good choice

Disk usage

- Low resolution - $N_{\text{vars}} \times N_r \times N_\theta \times N_\phi \times 8$ bytes $= 15 \times 128 \times 128 \times 256 \times 8 \approx \boxed{503.32\text{MB}(\Delta t)^{-1}}$
- High resolution - $N_{\text{vars}} \times N_r \times N_\theta \times N_\phi \times 8$ bytes $= 15 \times 128 \times 256 \times 512 \times 8 \approx \boxed{2.013\text{GB}(\Delta t)^{-1}}$
- Total (100ms or 20000M):

	20M	**50M**
Low resolution	≈ 503.32 GB	≈ 201.33 GB
High resolution	≈ 2013.64 GB	≈ 805.2 GB

GRMHD butterfly diagrams

Variables
Primitives: `bcc1`, `bcc2`, `bcc3`, `velx`, `vely`, `velz`
Gauge: `z4c_alpha`, `z4c_betax`, `z4c_betay`, `z4c_betaz`
Metric: `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, `adm_gzz`
Total: $\boxed{N_{\text{vars}} = 16}$

Output type

```
<output4>
file_type = sph
variable  = ...   # see variable list
dt        = 20.0
radius    = 400.0 # radii = 300.0 400.0 ...
```

Setup
Output time: 10M or 20M (the latter might already be too large)
Resolution: $(\theta, \phi)_{\text{low}} = (128, 256)$ or $(\theta, \phi)_{\text{high}} = (256, 512)$
Radii: Somewhere in the disk (between 50M and 300M)

Disk usage

- Low resolution - $N_{\text{vars}} \times N_\theta \times N_\phi \times 8$ bytes $= 16 \times 128 \times 256 \times 8 \approx \boxed{4.19\text{MB}(r\Delta t)^{-1}}$
- High resolution - $N_{\text{vars}} \times N_\theta \times N_\phi \times 8$ bytes $= 16 \times 256 \times 512 \times 8 \approx \boxed{16.78\text{MB}(r\Delta t)^{-1}}$
- Total (100ms or 20000M):

	10M	**20M**
Low resolution	≈ 8.38 GBr^{-1}	≈ 4.19 GBr^{-1}
High resolution	≈ 33.56 GBr^{-1}	≈ 16.78 GBr^{-1}

Note To save disk-usage, this can be paired with the output from **Ejecta** and also **Neutrino luminosity and average energy density**. The computation is performed via $b_{\bar{\phi}} = \frac{xb_y - yb_x}{\sqrt{x^2+y^2}}$, where b_x and b_y are the magnetic field components along the x- and y-direction in the fluid frame, and $\bar{\phi}$ denotes the average in the azimuthal direction.

Waveforms

Variables None.

Output type A block like this is needed inside of the `<z4c>` block:

```
nrad_wave_extraction = 5
extraction_radius_0  = 100
extraction_radius_1  = 200
extraction_radius_2  = 300
extraction_radius_3  = 400
extraction_radius_4  = 500
waveform_dt          = 4.0
```

This will generate two files (real/imag) for each radius with all the modes up to $l = 8$ by default.

Setup
Output time: 10M is a good minimum, but 20M might also be acceptable Radii: a wide range between 100M and 1000M is recommended for proper waveform extraction

Disk usage

- Bytes per line - $N_{\text{byte,column}} \times N_{\text{columns}} = \underbrace{21}_{\text{worstcase}} \times \underbrace{77}_{\text{for 1 passive scalar}} + \underbrace{13}_{\text{t-column}} \approx 1708$ Bytes per file
- Total (100ms or 20000M):

Num. radii /dt	**5M**	**10M**	**20M**
5	≈ 68.32 MB	≈ 34.16 MB	≈ 17.08 MB
10	≈ 136.64 MB	≈ 68.32 MB	≈ 34.16 MB

Kinetic/magnetic energy evolution

Variables None.

Output type

```
<output5>
file_type   = hst
dt          = 20.0
data_format = %20.15e
```

For this analysis we require the file `<jobname>.mhd.hst`.

Setup
Output time: 5M is a good minimum, but can also be increased to 10M or 20M, or lowered to 1M

Disk usage

- Bytes per line - $N_{\text{byte,column}} \times N_{\text{columns}} = \underbrace{22}_{\text{21 if unsigned}} \times \underbrace{14}_{\text{for 1 passive scalar}} \approx 294$ Bytes
- Total (100ms or 20000M):

	1M	5M	10M	20M
Usage	≈ 5.88 MB	≈ 1.18 MB	≈ 0.588 MB	≈ 0.294 MB

User history output

Variables None.

Output type

```
<output6>
file_type   = hst
dt          = 20.0
data_format = %20.15e
```

For this analysis we require the file `<jobname>.user.hst`. Note: the parameter `user_hist` inside of `<problem>` is needed for this output.

Setup
Output time: 5M is a good minimum, but can also be increased to 10M or 20M, or lowered to 1M

Disk usage

- Bytes per line - $N_{\text{byte,column}} \times N_{\text{columns}} = \underbrace{22}_{\text{21 if unsigned}} \times 4 \approx 88$ Bytes
- Total (100ms or 20000M):

	1M	5M	10M	20M
Usage	≈ 1.76 MB	≈ 0.352 MB	≈ 0.176 MB	≈ 0.088 MB

Spectra (MHD)

Variables
Primitives: `bcc1`, `bcc2`, `bcc3`, `velx`, `vely`, `velz`
Total: $\boxed{N_{\text{vars}} = 6}$

Output type

```
<output7>
file_type = bin
variable  = ... # see variable list
dt        = 20.0
```

Setup

Output time: 50M or 100M
Start: post-merger phase

Disk usage

- Usage per MeshBlock (32^3) - $N_{\text{vars}} \times N_{\text{points per MB}} = 6 \times 32^3 \times 8$ bytes $\approx$ $\boxed{\text{1.57MB per MeshBlock}}$
- Usage per MeshBlock (48^3) - $N_{\text{vars}} \times N_{\text{points per MB}} = 6 \times 48^3 \times 8$ bytes $\approx$ $\boxed{\text{5.31MB per MeshBlock}}$
- Usage per MeshBlock (64^3) - $N_{\text{vars}} \times N_{\text{points per MB}} = 6 \times 64^3 \times 8$ bytes $\approx$ $\boxed{\text{12.58MB per MeshBlock}}$
- Total (100ms or 20000M):

	50M	**100M**
$N_{\text{p.p.MB}} = 32^3;\ N_{\text{MB tot.}} = 1000$	$\approx$ 628 GB	$\approx$ 314 GB
$N_{\text{p.p.MB}} = 48^3;\ N_{\text{MB tot.}} = 1000$	$\approx$ 2.124 TB	$\approx$ 1.062 TB
$N_{\text{p.p.MB}} = 64^3;\ N_{\text{MB tot.}} = 1000$	$\approx$ 5.032 TB	$\approx$ 2.516 TB